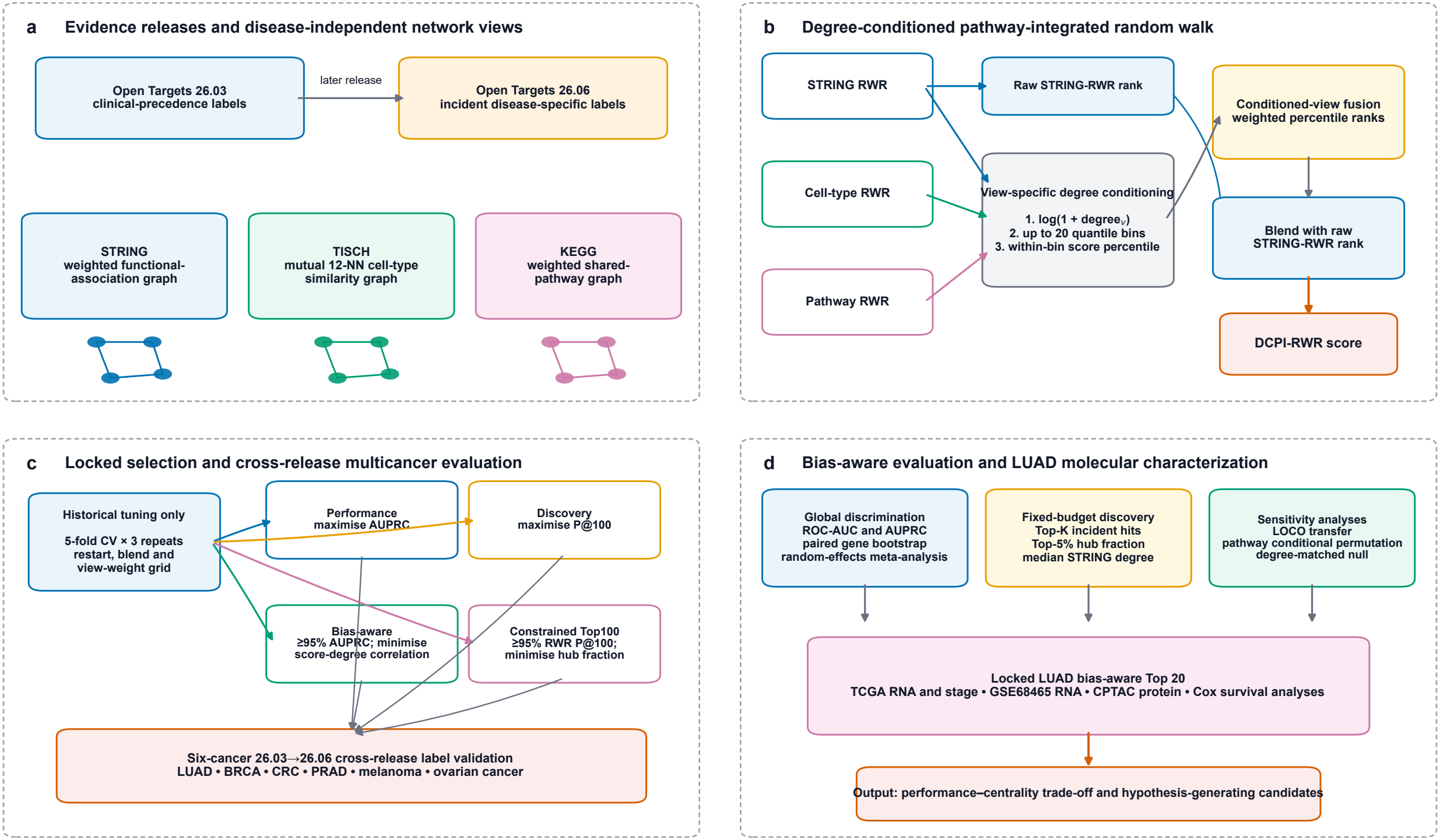
a
Evidence releases and disease-independent network views
Open Targets 26.03
clinical-precedence labels
later release
Open Targets 26.06
incident disease-specific labels
STRING
weighted functional-association graph
TISCH
mutual 12-NN cell-type similarity graph
KEGG
weighted shared-pathway graph
b
Degree-conditioned pathway-integrated random walk
STRING RWR
Raw STRING-RWR rank
Conditioned-view fusion
weighted percentile ranks
Cell-type RWR
View-specific degree conditioning
1. log(1 + degree$_v$)
2. up to 20 quantile bins
3. within-bin score percentile
Blend with raw
STRING-RWR rank
Pathway RWR
DCPI-RWR score
c
Locked selection and cross-release multicancer evaluation
Historical tuning only
5-fold CV × 3 repeats
restart, blend and
view-weight grid
Performance
maximise AUPRC
Discovery
maximise P@100
Bias-aware
≥95% AUPRC; minimise
score-degree correlation
Constrained Top100
≥95% RWR P@100;
minimise hub fraction
Six-cancer 26.03→26.06 cross-release label validation
LUAD • BRCA • CRC • PRAD • melanoma • ovarian cancer
d
Bias-aware evaluation and LUAD molecular characterization
Global discrimination
ROC-AUC and AUPRC
paired gene bootstrap
random-effects meta-analysis
Fixed-budget discovery
Top-K incident hits
Top-5% hub fraction
median STRING degree
Sensitivity analyses
LOCO transfer
pathway conditional permutation
degree-matched null
Locked LUAD bias-aware Top 20
TCGA RNA and stage • GSE68465 RNA • CPTAC protein • Cox survival analyses
Output: performance–centrality trade-off and hypothesis-generating candidates